Nonconcavity of Robin Eigenvalues on Convex Trapezoids

An affine-width counterexample to Laugesen's Conjecture C

Keio Research Notes

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Abstract

We construct bounded convex planar domains for which the second Robin eigenvalue is not concave in the positive boundary parameter, giving a negative answer to Conjecture C of Laugesen (2019). The domains are thin trapezoids of width $2\varepsilon(1+\delta x)$. With Robin parameter $\alpha = \varepsilon t$, their low eigenvalues converge to a weighted Neumann problem on the unit interval. For its $(n+1)$st eigenvalue, $n \geq 1$, we prove the exact response

$$\partial_t^2 \Lambda_{n+1}(\delta, t) = \frac{n^2\pi^2 + 21}{24 n^4 \pi^4}\, \delta^2 + O(\delta^3),$$

uniformly on each fixed compact t-interval. The coefficient is positive. A strict midpoint defect at $t = 1, 2, 3$ therefore transfers to the original Robin problem at $\alpha = \varepsilon, 2\varepsilon, 3\varepsilon$ on one fixed convex domain. For every prescribed finite M, the same domain and parameters work simultaneously for the eigenvalues $\lambda_2, \ldots, \lambda_M$. We also establish a general width-profile response formula, stability under nearby concave profiles, and extensions to real-analytic strictly convex domains in every dimension at least two. The proofs are analytic and use neither an unproved conjecture nor a numerical sign certificate. Variational convergence and eigenvalue perturbation are established tools; the contribution is their explicit affine-width application to the stated concavity problem.

1 The original problem and the main result

For a bounded Lipschitz domain $\Omega \subset \mathbb{R}^d$, let $\lambda_j(\Omega;\alpha)$ be the Robin eigenvalues, counted with multiplicity and starting at $j = 1$, with outward-normal convention

$$-\Delta u = \lambda u \quad \text{in } \Omega, \qquad \partial_\nu u + \alpha u = 0 \quad \text{on } \partial\Omega. \tag{1}$$

The scalar parameter α is constant on the entire boundary. Equivalently these are the eigenvalues of the closed quadratic form

$$q_{\Omega,\alpha}[u] = \int_\Omega |\nabla u|^2 \, dx + \alpha \int_{\partial\Omega} |u|^2 \, dS, \qquad u \in H^1(\Omega), \tag{2}$$

in $L^2(\Omega)$. The first eigenvalue is concave in α by the Rayleigh principle. The question for the second eigenvalue is different because its variational characterization is a min–max.

Laugesen's Conjecture C [1, Section 2, p. 7] asserts that every bounded convex Ω has

$$\alpha \longmapsto \lambda_2(\Omega;\alpha) \quad \text{concave on } (0,\infty). \tag{3}$$

It is also listed in the 2020 eigenvalue miniconference [2, Open Problem 7] and in the 2023 Oberwolfach report [3, p. 2099]. The result below concerns this original positive-parameter, convex-domain statement.

For $\varepsilon, \delta > 0$, set

$$\Omega_{\varepsilon,\delta} = \{(x, y) \in \mathbb{R}^2 : 0 < x < 1,\ |y| < \varepsilon(1 + \delta x)\}. \tag{4}$$

This is the intersection of four open half-planes, hence is a bounded open convex trapezoid with Lipschitz boundary. Define its midpoint defect by

$$D_j(\varepsilon, \delta) = \lambda_j(\Omega_{\varepsilon,\delta}; \varepsilon) + \lambda_j(\Omega_{\varepsilon,\delta}; 3\varepsilon) - 2\lambda_j(\Omega_{\varepsilon,\delta}; 2\varepsilon). \tag{5}$$

A concave eigenvalue curve would require $D_j \le 0$.

Theorem 1.1 (Convex-domain nonconcavity)**.** *For every integer $M \ge 2$ there is a number $\delta_M > 0$ such that for every $\delta \in (0, \delta_M)$ there exists $\varepsilon_0(\delta, M) > 0$ for which*

$$D_j(\varepsilon, \delta) > 0 \qquad (0 < \varepsilon < \varepsilon_0(\delta, M),\ 2 \le j \le M). \tag{6}$$

In particular, Conjecture (3) is false. For each fixed $j \ge 2$, the precise iterated limit is

$$\lim_{\delta \downarrow 0} \frac{1}{\delta^2} \lim_{\varepsilon \downarrow 0} D_j(\varepsilon, \delta) = \frac{(j-1)^2\pi^2 + 21}{24(j-1)^4\pi^4} > 0. \tag{7}$$

All parameters are positive. After selecting δ and ε, the three eigenvalues in (5) are evaluated on the same domain. The two limits in (7) are taken in the displayed order; no estimate uniform in all j is asserted. The proof establishes an open range of sufficiently small positive parameters, rather than an unverified decimal example.

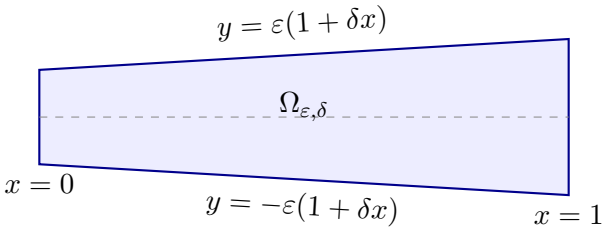


Figure 1: The convex trapezoid. Vertical scale is exaggerated. The same Robin parameter is imposed on all four sides.

1.1 How the proof is organized

The argument has three ingredients. First, the scaling $\alpha = \varepsilon t$ converts the boundary term of a thin domain into an effective multiplication operator on an interval. Second, a reduced-resolvent identity isolates the exact quadratic effect of changing the width. Third, a finite midpoint inequality passes through eigenvalue convergence. This last step avoids differentiating a thin-domain limit.

The main construction uses only the affine profile $x \mapsto x$. We first prove a response formula for general Lipschitz profiles to identify which part of the calculation controls the sign. The full proofs include the change in the weighted Hilbert space, all boundary faces, and the ordering of eigenvalues.

1.2 Provenance and scope

The source problem is independent of the supplied Verifiable Mathematical Reasoning Corpus [8]. We checked the 95 TeX manuscripts in its 29 October 2026 snapshot and found no instance

of the present positive-parameter concavity counterexample. The corpus and earlier research notes motivated the search for exact finite certificates and for transfers that preserve the original quantifiers. The present proof does not assume any unverified corpus theorem or any arithmetic statement about Euler's constant.

The mathematical theorem and a claim of historical priority are separate matters. Section 10 records the primary sources examined and the limits of the novelty audit. This is a research manuscript prepared with AI assistance, not a report of external peer review.

2 Thin convex domains and an effective boundary potential

For a positive Lipschitz function $h : [0,1] \to (0,\infty)$ and $\varepsilon > 0$, put

$$\Omega_{\varepsilon,h} = \{(x,y) : 0 < x < 1,\ -\varepsilon h(x) < y < \varepsilon h(x)\}.$$

If h is concave, $\Omega_{\varepsilon,h}$ is convex. Its Robin eigenvalues $\lambda_j(\Omega_{\varepsilon,h};\alpha)$ correspond to $\partial_\nu u + \alpha u = 0$ on the whole boundary and are numbered starting at $j = 1$. Define $\Lambda_j(h,t)$ by the regular weighted Sturm–Liouville form

$$a_{h,t}[u] = \int_0^1 h(x)|u'(x)|^2\,dx + t\int_0^1 |u(x)|^2\,dx, \qquad u \in H^1(0,1), \tag{8}$$

in $L^2((0,1), h\,dx)$. Thus the endpoint conditions are Neumann and the equation is

$$-(hu')' + tu = \Lambda h u, \qquad u'(0) = u'(1) = 0.$$

Lemma 2.1 (Thin-domain convergence)**.** *For every positive Lipschitz h, every $t \ge 0$, and every fixed $j \ge 1$,*

$$\lim_{\varepsilon \downarrow 0} \lambda_j(\Omega_{\varepsilon,h};\varepsilon t) = \Lambda_j(h,t). \tag{9}$$

Proof. On $D = (0,1)\times(-1,1)$ make the change of variables $F_\varepsilon(x,s) = (x, \varepsilon h(x)s)$ and write $v = u \circ F_\varepsilon$. After division of the numerator and denominator of the Rayleigh quotient by ε, the squared norm becomes

$$m[v] = \int_D h(x)|v(x,s)|^2\,dx\,ds,$$

and its numerator is exactly

$$\begin{aligned} q_{\varepsilon,t}[v] = {} & \int_D h\left|v_x - \frac{h'}{h}s v_s\right|^2 dx\,ds + \varepsilon^{-2}\int_D h^{-1}|v_s|^2\,dx\,ds \\ & + t\int_0^1 \sqrt{1+\varepsilon^2 h'(x)^2}\,(|v(x,1)|^2 + |v(x,-1)|^2)\,dx \\ & + \varepsilon t \sum_{r\in\{0,1\}} h(r)\int_{-1}^1 |v(r,s)|^2\,ds. \end{aligned} \tag{10}$$

Every term is nonnegative. For functions independent of s, $m[v] = 2\int h|u|^2$ and $q_{\varepsilon,t}[v] \to 2a_{h,t}[u]$. The convergence is uniform on each fixed finite-dimensional subspace of $H^1(0,1)$. Applying the minimax principle to the span of the first j effective eigenfunctions proves the upper bound in (9).

For the lower bound, take a sequence along which the lower limit of the jth eigenvalue is attained, and choose m-orthonormal eigenfunctions $v_{\varepsilon,1},\dots,v_{\varepsilon,j}$ for the first j eigenvalues. The upper bound controls their forms. The second term in (10) gives $\|\partial_s v_{\varepsilon,k}\|_{L^2(D)} = O(\varepsilon)$. Since h is bounded above and below and h' is bounded, the first term then bounds $\partial_x v_{\varepsilon,k}$ in $L^2(D)$. Consequently these functions are bounded in $H^1(D)$. After a subsequence they converge weakly

in $H^1(D)$, strongly in $L^2(D)$, and their limits are independent of s and remain m-orthonormal. To justify the boundary terms directly, put $\bar{v}(x) = \frac{1}{2}\int_{-1}^{1} v(x,s)\,ds$. For almost every x,

$$|v(x,\pm 1) - \bar{v}(x)|^2 \le 2\int_{-1}^{1} |v_s(x,s)|^2\,ds.$$

Thus the two sloping-side traces converge strongly in $L^2(0,1)$ to the same limiting longitudinal function. The endpoint boundary terms need no convergence assertion for the lower bound.

For every fixed linear combination of these j functions, weak lower semicontinuity applies to the longitudinal term; the top and bottom traces converge strongly; and the transverse and endpoint terms may be discarded. The corresponding effective Rayleigh quotient is therefore at most the lower limit of the jth eigenvalue. The limit functions span a j-dimensional space, so a second application of minimax proves the reverse bound. □

3 A curvature response for width profiles

Let $f \in W^{1,\infty}(0,1)$ be real, set $h_\delta = 1 + \delta f$, and restrict δ to a neighborhood of zero where $h_\delta > 0$. Write $\Lambda_j(\delta,t) = \Lambda_j(h_\delta,t)$. On unweighted $L^2(0,1)$, let

$$E_n = n^2\pi^2, \qquad \phi_0 = 1, \qquad \phi_n(x) = \sqrt{2}\cos(n\pi x) \quad (n \ge 1).$$

These are the eigenpairs of the Neumann Laplacian A_0. Let P_n be the orthogonal projection onto ϕ_n and define the bounded reduced resolvent

$$R_n = (1-P_n)(E_n - A_0)^{-1}(1-P_n),$$

where the inverse is taken only on the orthogonal complement. For a real bounded function f, define

$$\mathcal{Q}_n(f) = 2\langle f\phi_n, R_n f\phi_n\rangle = 2\sum_{m\ne n} \frac{|\langle f\phi_n, \phi_m\rangle|^2}{E_n - E_m}. \tag{11}$$

The series is absolutely convergent: the reduced resolvent is bounded and the squared Fourier coefficients are summable.

An auxiliary design question. Can one choose a single nonconstant concave profile f such that $\mathcal{Q}_n(f) > 0$ for every $n \ge 1$, while $\mathcal{Q}_0(f) < 0$? This is a design problem for the limiting response, introduced here to find counterexamples to the original Robin conjecture. Lemma 4.1 answers it with the affine profile $f(x) = x$. The all-index assertion concerns the response coefficients; transfer to one physical domain below is asserted only for a prescribed finite set of eigenvalues.

Lemma 3.1 (Simple-eigenvalue curvature)**.** *Let $H(s) = H_* + sC$ be a self-adjoint family with compact resolvent, where C is bounded and self-adjoint. Suppose $e(s)$ is a simple eigenvalue with normalized real eigenvector $\psi(s)$. Write P for its eigenprojection and*

$$R = (1-P)(e(s) - H(s))^{-1}(1-P)$$

for the inverse on the orthogonal complement. Then

$$e''(s) = 2\langle C\psi(s), RC\psi(s)\rangle. \tag{12}$$

Replacing C by $C - cI$ on the right does not change the result.

Proof. Choose the real normalization with $\langle \psi, \psi' \rangle = 0$. Differentiating the eigenvalue equation gives

$$e' = \langle C\psi, \psi \rangle, \qquad (e - H)\psi' = (C - e')\psi.$$

The latter identity on the orthogonal complement yields $\psi' = RC\psi$. Differentiating the first identity proves (12). Since $R\psi = 0$ and R is self-adjoint, scalar multiples of the identity disappear. Existence of the derivatives follows from the isolated simple eigenvalue and the resolvent Neumann series; this is the standard simple-eigenvalue perturbation argument [4, Chapter VII]. □

Lemma 3.2 (Uniform width-profile curvature)**.** *Let $f \in W^{1,\infty}(0,1)$ be real, let $h_\delta = 1 + \delta f$, and let $\Lambda_j(\delta, t)$ be the eigenvalues of (8) with this weight. For every $T > 0$ and integer $N \geq 0$,*

$$\lim_{\delta \to 0} \max_{0 \leq n \leq N} \sup_{0 \leq t \leq T} \left| \delta^{-2} \partial_t^2 \Lambda_{n+1}(\delta, t) - \mathcal{Q}_n(f) \right| = 0, \tag{13}$$

where $\mathcal{Q}_n$ is defined by (11). More precisely, for these fixed T, N, f, the supremum in (13) is $O(|\delta|)$; hence

$$\partial_t^2 \Lambda_{n+1}(\delta, t) = \delta^2 \mathcal{Q}_n(f) + O(\delta^3) \quad (0 \leq t \leq T,\ 0 \leq n \leq N). \tag{14}$$

The derivatives in this statement exist for every sufficiently small $|\delta|$, uniformly on the indicated compact interval.

Proof. Under the unitary map $u \mapsto \sqrt{h_\delta}\, u$, the effective operator is $A_\delta + tV_\delta$, where A_δ is associated with the common-domain form

$$b_\delta[g] = \int_0^1 \left| g' - \frac{\delta f'}{2(1 + \delta f)} g \right|^2 dx, \qquad g \in H^1(0,1),$$

and V_δ is multiplication by $(1 + \delta f)^{-1}$. Define

$$W_\delta = \frac{V_\delta - I}{\delta} = -\frac{f}{1 + \delta f}, \qquad B_{\delta,t} = A_\delta + t(V_\delta - I).$$

The eigenvalues of $B_{\delta,t}$ are $\Lambda_j(\delta, t) - t$. This subtraction leaves all spectral gaps and all second derivatives in t unchanged.

The forms of $B_{\delta,t}$ converge to the Neumann form $b_0[g] = \int |g'|^2$ in form norm, uniformly for $0 \leq t \leq T$. Indeed, their difference is bounded in absolute value by $C_T|\delta|\|g\|_{H^1}^2$. After adding a fixed sufficiently large positive multiple of $\|g\|_{L^2}^2$, these forms are uniformly coercive. The variational resolvent identity then gives norm-resolvent convergence uniform in t, with error $O(|\delta|)$. Explicitly, regard the shifted forms as bounded invertible maps $H^1 \to (H^1)^*$ and apply $L_\delta^{-1} - L_0^{-1} = L_\delta^{-1}(L_0 - L_\delta)L_0^{-1}$. The continuous embeddings from L^2 to $(H^1)^*$ and from H^1 to L^2 give the claimed operator-norm estimate. The resolvent identity propagates it to any fixed compact contour in the resolvent set of A_0. Since the finitely many Neumann eigenvalues $E_0, \ldots, E_N$ are isolated and simple, fixed disjoint small circles around them give spectral projections converging in norm, uniformly in t. In particular, the corresponding normalized real eigenfunctions $\psi_{n,\delta,t}$ may be chosen with

$$\sup_{0 \leq t \leq T} \|\psi_{n,\delta,t} - \phi_n\|_{L^2} \to 0.$$

Their reduced resolvents $R_{n,\delta,t}$ converge in operator norm to R_n, also uniformly in t. All of these convergences have error $O(|\delta|)$. For the reduced resolvent this follows, for example, from

$$R_{n,\delta,t} = \frac{1}{2\pi i} \int_{\Gamma_n} \frac{(z - B_{\delta,t})^{-1}}{z - e_{n,\delta,t}}\, dz, \qquad e_{n,\delta,t} = \Lambda_{n+1}(\delta, t) - t,$$

where Γ_n is a fixed small positively oriented circle enclosing only this eigenvalue. The eigenprojection contribution has a double pole of zero contour integral; every other spectral component

has residue $(e_{n,\delta,t} - e_m)^{-1}$. Simplicity on a neighborhood of the compact parameter interval and the common analytic form domain justify ordinary simple-eigenvalue differentiation.

Because $\partial_t B_{\delta,t} = V_\delta - I = \delta W_\delta$, Lemma 3.1 gives the exact identity

$$\delta^{-2}\partial_t^2 \Lambda_{n+1}(\delta, t) = 2\langle W_\delta \psi_{n,\delta,t}, R_{n,\delta,t} W_\delta \psi_{n,\delta,t}\rangle.$$

Finally $W_\delta \to -f$ in multiplication-operator norm. The uniform convergence, with error $O(|\delta|)$, of all three factors on the right proves (13). Taking a maximum over finitely many n does not affect convergence. □

4 An exact calculation for affine width

Lemma 4.1 (Linear-profile response)**.** *For $f(x) = x$,*

$$\mathcal{Q}_0(x) = -\frac{1}{60}, \qquad \mathcal{Q}_n(x) = \frac{n^2\pi^2 + 21}{24n^4\pi^4} > 0 \quad (n \geq 1). \tag{15}$$

Proof. Fix $n \geq 1$, put $k = n\pi$, and define

$$y_n(x) = \sqrt{2}\left[\left(\frac{x^2 - x}{4k} - \frac{1}{4k^3}\right)\sin(kx) + \frac{x - 1/2}{4k^2}\cos(kx)\right].$$

Direct differentiation gives

$$y_n'' + k^2 y_n = (x - 1/2)\phi_n, \quad y_n'(0) = y_n'(1) = 0, \quad \int_0^1 y_n \phi_n \, dx = 0.$$

Since $\int_0^1 x\phi_n^2 \, dx = 1/2$, these identities characterize $y_n = R_n x\phi_n$. Integration by parts gives

$$\int_0^1 x(x^2 - x)\cos(kx)\sin(kx)\, dx = \frac{3}{8k^3},$$
$$\int_0^1 x(x - 1/2)\cos^2(kx)\, dx = \frac{1}{24} + \frac{1}{4k^2},$$
$$\int_0^1 x\cos(kx)\sin(kx)\, dx = -\frac{1}{4k}.$$

It follows that

$$\langle x\phi_n, y_n\rangle = \frac{1}{48k^2} + \frac{7}{16k^4},$$

which proves the second formula in (15). For $n = 0$, the corresponding correction is $y_0(x) = x^3/6 - x^2/4 + 1/24$; it has mean zero and satisfies $y_0'' = x - 1/2$ and the Neumann endpoint conditions. Hence $\mathcal{Q}_0(x) = 2\int_0^1 xy_0 \, dx = -1/60$. □

5 A fixed finite defect and the original conjecture

Lemma 5.1 (A fixed three-point defect)**.** *For every fixed $n \geq 0$ and real $f \in W^{1,\infty}(0, 1)$,*

$$\lim_{\delta \to 0} \frac{\Lambda_{n+1}(\delta, 1) + \Lambda_{n+1}(\delta, 3) - 2\Lambda_{n+1}(\delta, 2)}{\delta^2} = \mathcal{Q}_n(f). \tag{16}$$

Proof. For every twice continuously differentiable F on $[1, 3]$,

$$F(1) + F(3) - 2F(2) = \int_1^2 (t - 1)F''(t)\, dt + \int_2^3 (3 - t)F''(t)\, dt.$$

The two nonnegative weights have total integral one. Apply Lemma 3.2 with $T = 3$. □

Theorem 5.2 (Convex trapezoids with fixed parameter ratios)**.** *For every integer $M \geq 2$, put*

$$c_M = \frac{(M-1)^2\pi^2 + 21}{24(M-1)^4\pi^4} > 0.$$

There is $\delta_M \in (0,1)$ such that, for every $\delta \in (0,\delta_M)$, there is $\varepsilon_M(\delta) > 0$ for which every $0 < \varepsilon < \varepsilon_M(\delta)$ satisfies

$$\begin{aligned} &\lambda_j(\Omega_{\varepsilon,\delta};\varepsilon) + \lambda_j(\Omega_{\varepsilon,\delta};3\varepsilon) - 2\lambda_j(\Omega_{\varepsilon,\delta};2\varepsilon) > \frac{1}{4}c_M\delta^2, \quad 2 \leq j \leq M, \\ &\Omega_{\varepsilon,\delta} = \{(x,y) : 0 < x < 1,\ |y| < \varepsilon(1+\delta x)\}. \end{aligned} \tag{17}$$

Consequently one fixed convex planar domain simultaneously violates concavity of the first $M-1$ excited Robin eigenvalues, at the same three positive parameters $\varepsilon, 2\varepsilon, 3\varepsilon$.

Proof. Take $f(x) = x$. The exact formula (15) decreases with $n \geq 1$, so $c_M = \min_{1\leq n<M} \mathcal{Q}_n(x)$. By Lemma 5.1, a common $\delta_M > 0$ can be chosen so that every effective defect for $j = 2,\dots,M$ is greater than $c_M\delta^2/2$, whenever $0 < \delta < \delta_M$. Fix such a δ. Lemma 2.1 applies at the finitely many pairs (j,t) with $2 \leq j \leq M$ and $t \in \{1,2,3\}$. For sufficiently small ε, every eigenvalue error there is less than $c_M\delta^2/16$. The error of a three-point defect is then less than $c_M\delta^2/4$, proving (17). □

Completion of the proof of Theorem 1.1. The strict inequality (6) follows from Theorem 5.2. For each fixed δ, Lemma 2.1 identifies the inner limit in (7) with the effective three-point defect. Lemma 5.1 and Lemma 4.1 then give the stated outer limit. For $j = 2$, the resulting strict defect contradicts the midpoint inequality required by (3). □

Remark 5.3 (Why the limit order is sufficient). Let $\Delta_j(\delta)$ denote the effective defect. For a chosen δ the proof establishes $\Delta_j(\delta) > 0$. Choose ε only after this step, so that each of the three eigenvalue errors is less than $\Delta_j(\delta)/8$. The error of the defect is then less than $\Delta_j(\delta)/2$. This is a finite convergence argument, not an exchange of derivatives or limits. The explicit positive lower bound (17) uses a common version of the same estimate for finitely many indices.

6 Which width changes produce the positive sign?

The response formula separates the width calculation from the thin-domain limit. Its sign is an indefinite spectral quadratic form, rather than an automatic consequence of positivity of the boundary form.

Proposition 6.1 (Affine profiles and reflection-symmetric profiles)**.** *For every $n \geq 0$ and real constants a, b,*

$$\mathcal{Q}_n(a + bx) = b^2\mathcal{Q}_n(x).$$

If $f \in L^\infty(0,1)$ is real, not almost everywhere constant, and $f(1-x) = f(x)$ almost everywhere, then

$$\mathcal{Q}_1(f) < 0.$$

For every nonconstant real bounded profile, $\mathcal{Q}_0(f) < 0$.

Proof. Since $R_n\phi_n = 0$, addition of a constant to f does not change $\mathcal{Q}_n$, and homogeneity gives the affine identity. For $n = 0$ every nonzero summand has a negative denominator. Equality would force $f\phi_0$ to lie in the span of ϕ_0, which means that f is constant.

For the second assertion, ϕ_1 is odd under $x \mapsto 1-x$. Thus $f\phi_1$ is odd and is orthogonal to ϕ_0, as well as to all even-index cosine modes. The only positive-denominator term in $\mathcal{Q}_1$ is the

ϕ_0 term, which vanishes. All remaining terms, other than the omitted ϕ_1 term, have negative denominators. If they all vanished, completeness of the cosine basis would give $f\phi_1 = c\phi_1$ almost everywhere. Since ϕ_1 is nonzero except at one point, this again forces f to be constant. □

The affine width breaks reflection symmetry in the longitudinal coordinate. The trapezoid still has a transverse reflection symmetry, but its low excited modes are longitudinal; that symmetry does not remove their coupling to the ground mode. This is consistent with the symmetry mechanism discussed in [1].

Proposition 6.2 (An explicit neighborhood of useful profiles)**.** *For $M \geq 2$, put*

$$q_n = \frac{n^2\pi^2 + 21}{24n^4\pi^4}, \qquad g_n = (2n-1)\pi^2 \quad (n \geq 1), \qquad \eta_M = \min\left\{1, \min_{1\leq n<M} \frac{g_n q_n}{12}\right\}.$$

If f is real and $\|f - x\|_\infty < \eta_M$, then $\mathcal{Q}_n(f) > q_n/2$ for $1 \leq n < M$. Consequently every concave $f \in W^{1,\infty}(0,1)$ in this neighborhood produces the simultaneous concavity violations of Theorem 1.1, on the convex domains $\Omega_{\varepsilon,1+\delta f}$, for sufficiently small $\delta > 0$ and then sufficiently small $\varepsilon > 0$.

Proof. The spectral distance of E_n from the other Neumann eigenvalues is g_n, so $\|R_n\| = 1/g_n$. Expanding the difference of the two quadratic forms gives

$$|\mathcal{Q}_n(f) - \mathcal{Q}_n(x)| \leq \frac{2}{g_n}\|f - x\|_\infty(\|f\|_\infty + \|x\|_\infty) \leq \frac{6}{g_n}\|f - x\|_\infty.$$

The chosen neighborhood makes the last quantity less than $q_n/2$. Use Lemma 5.1 for these finitely many positive responses and then Lemma 2.1. Concavity of f and positivity of $1 + \delta f$ give convexity of the domain. The threshold for δ may depend on f, including its Lipschitz constant. □

7 Higher-dimensional convex polytopes

Corollary 7.1 (Every ambient dimension)**.** *For every $d \geq 2$ and $M \geq 2$, there are bounded convex polytopes in $\mathbb{R}^d$ and positive parameters $\alpha, 2\alpha, 3\alpha$ for which*

$$\lambda_j(\Omega;\alpha) + \lambda_j(\Omega;3\alpha) > 2\lambda_j(\Omega;2\alpha), \qquad 2 \leq j \leq M.$$

One may use the explicit family

$$\Omega_{\varepsilon,\delta}^{(d,M)} = \Omega_{\varepsilon,\delta} \times (0, 1/(2M))^{d-2}, \qquad \alpha = \varepsilon, \tag{18}$$

with $\delta > 0$ sufficiently small and then $\varepsilon > 0$ sufficiently small. The empty Cartesian factor for $d = 2$ is understood.

Proof. The planar assertion has already been proved. Fix $d \geq 3$ and write $\ell = 1/(2M)$. Let $\eta_k(\alpha)$ denote the Robin eigenvalues of $(0,\ell)$, starting with $k = 1$. The Robin form on a Cartesian product separates, so its complete spectrum is formed by sums of the spectra of the factors. Since

$$\eta_1(\varepsilon t) \to 0, \qquad \eta_2(\varepsilon t) - \eta_1(\varepsilon t) \to \pi^2/\ell^2 = 4M^2\pi^2$$

for each fixed $t \geq 0$, excitation of an extra interval factor has a uniformly large limiting cost at $t \in \{1,2,3\}$.

To check that the first M eigenvalues arise from planar excitations, take $0 < \delta < 1$. Directly from the effective Rayleigh quotient and $1 \leq h_\delta \leq 1 + \delta$,

$$\frac{(j-1)^2\pi^2 + t}{1+\delta} \leq \Lambda_j(\delta, t) \leq (1+\delta)(j-1)^2\pi^2 + t.$$

Hence for $1 \le t \le 3$,

$$\Lambda_M(\delta, t) - \Lambda_1(\delta, t) \le 2(M-1)^2\pi^2 + \frac{3}{2} < 4M^2\pi^2.$$

The planar thin-domain limit preserves this strict separation for all sufficiently small ε, simultaneously at $t = 1, 2, 3$. Thus for $1 \le j \le M$ at these parameters,

$$\lambda_j(\Omega_{\varepsilon,\delta}^{(d,M)}; \varepsilon t) = \lambda_j(\Omega_{\varepsilon,\delta}; \varepsilon t) + (d-2)\eta_1(\varepsilon t).$$

For fixed d, the extra term tends to zero. Consequently each product-domain midpoint defect has exactly the same positive limit as its planar effective defect. Choose δ as in Theorem 5.2, and then take ε sufficiently small. This proves the assertion on the fixed convex polytope (18). □

8 Real-analytic strictly convex boundaries

Lemma 8.1 (Smooth strictly convex approximation and Robin continuity)**.** *Let $P \subset \mathbb{R}^d$ be a bounded open convex polytope with nonempty interior. There are bounded domains P_η with real-analytic boundary and strictly positive principal curvatures, for $\eta > 0$ sufficiently small, such that $\overline{P_\eta} \to \overline{P}$ in Hausdorff distance and*

$$\lambda_j(P_\eta; \alpha) \longrightarrow \lambda_j(P; \alpha) \qquad (\eta \downarrow 0)$$

for every fixed $j \ge 1$ and $\alpha \ge 0$.

Proof. After a translation suppose $B(0, r) \subset P \subset B(0, R)$, where $r > 0$, and write

$$P = \{x : \ell_i(x) < 1 \text{ for } 1 \le i \le m\}, \qquad p(x) = \max_i \ell_i(x),$$

with linear functions $\ell_i(x) = a_i \cdot x$. Redundant copies of facets may be removed. Define

$$F_\eta(x) = \eta \log\left(\sum_{i=1}^m \exp(\ell_i(x)/\eta)\right) + \eta|x|^2, \qquad P_\eta = \{x : F_\eta(x) < 1\}.$$

The Hessian of the logarithmic term is positive semidefinite, and $\nabla^2 F_\eta \ge 2\eta I$. Also

$$p(x) \le F_\eta(x) \le p(x) + \eta \log m + \eta|x|^2.$$

Consequently $P_\eta \subset P$, $F_\eta(0) = \eta \log m < 1$ for small η, and the sets converge to P in Hausdorff distance. The level set $F_\eta = 1$ is regular: a critical point of a differentiable convex function is a global minimizer, whereas the minimum here is below 1. Its second fundamental form, with the outward orientation, is $\nabla^2 F_\eta$ restricted to its tangent space divided by $|\nabla F_\eta|$. Since F_η is real analytic, the regular-level-set theorem also gives real-analytic boundary. All its principal curvatures are positive.

Here are the convergence details needed for the boundary term in the Robin form. Let ρ and ρ_η be the radial functions of P and P_η on the unit sphere. Since $\rho(\omega) = 1/p(\omega)$ and $r \le \rho \le R$, the displayed bounds imply $\rho_\eta \to \rho$ uniformly, and $r/2 \le \rho_\eta \le R$ for sufficiently small η. Convexity gives, at $x = \rho_\eta(\omega)\omega$,

$$\nabla F_\eta(x) \cdot x \ge F_\eta(x) - F_\eta(0) = 1 - \eta \log m \ge \tfrac{1}{2}.$$

Moreover $|\nabla F_\eta| \le \max_i |a_i| + 2\eta R$ on $\overline{P}$. The implicit derivative formula

$$\nabla_S \rho_\eta(\omega) = -\rho_\eta(\omega) \frac{(I - \omega \otimes \omega)\nabla F_\eta(\rho_\eta \omega)}{\nabla F_\eta(\rho_\eta \omega) \cdot \omega}$$

therefore bounds the spherical gradients uniformly in η. For almost every ω, a unique facet maximizes $\ell_i(\omega)$: ties lie in finitely many great subspheres, of sphere measure zero. At each such ω the softmax weights defining ∇F_η converge to the unique maximizing facet. Hence

$$\nabla_S \rho_\eta \longrightarrow \nabla_S \rho \quad \text{almost everywhere on the sphere.}$$

This is almost-everywhere convergence with a uniform bound, not uniform convergence of derivatives across the corners.

Set $a_\eta = \rho_\eta/\rho$, and define the radial map

$$T_\eta(x) = a_\eta(x/|x|)x \quad (x \neq 0), \qquad T_\eta(0) = 0.$$

It maps P onto P_η. The bounds just obtained show that T_η and its inverse are Lipschitz with constants bounded independently of small η. At almost every $x = s\omega$,

$$DT_\eta = a_\eta I + \omega \otimes \nabla_S a_\eta, \qquad J_\eta := \det DT_\eta = a_\eta^d.$$

Thus $DT_\eta \to I$ almost everywhere, while $J_\eta \to 1$ uniformly. The volume energy matrix

$$A_\eta = J_\eta (DT_\eta)^{-1}(DT_\eta)^{-T}$$

converges almost everywhere to I and is uniformly bounded and uniformly positive definite. The surface Jacobian, expressed on ∂P using its radial parametrization, is

$$b_\eta(\rho(\omega)\omega) = \frac{\rho_\eta(\omega)^{d-1}\sqrt{1 + |\nabla_S \log \rho_\eta(\omega)|^2}}{\rho(\omega)^{d-1}\sqrt{1 + |\nabla_S \log \rho(\omega)|^2}}.$$

It tends to 1 almost everywhere and is bounded above and below by positive constants independent of small η.

Pulling the Robin form on P_η back by T_η now gives, on the fixed space $H^1(P)$,

$$a_\eta[u] = \int_P \nabla u \cdot A_\eta \nabla u + \alpha \int_{\partial P} b_\eta |u|^2, \qquad m_\eta[u] = \int_P J_\eta |u|^2.$$

For each fixed $u \in H^1(P)$ these converge to the usual Robin energy and mass on P, by dominated convergence. The convergence is uniform on the unit sphere of any fixed finite-dimensional subspace. The min–max principle therefore supplies the eigenvalue upper bounds.

For the lower bounds take first j eigenfunctions, pulled back to P and orthonormal for m_η. The upper bounds, uniform ellipticity, and $\alpha \geq 0$ make them bounded in $H^1(P)$. A common subsequence converges weakly in $H^1(P)$ and strongly in $L^2(P)$. Their traces converge strongly in $L^2(\partial P)$ as well. One can see the last fact directly here, without assuming uniform convergence at the corners: since $x \cdot \nu \geq r$ on every facet of P, the divergence theorem and density in $H^1(P)$ give

$$r\|w\|^2_{L^2(\partial P)} \leq d\|w\|^2_{L^2(P)} + 2R\|w\|_{L^2(P)}\|\nabla w\|_{L^2(P)}.$$

Apply this to each difference between an eigenfunction and its limit. The uniform bound on b_η and dominated convergence then give convergence of the Robin boundary terms.

If $u_\eta \rightharpoonup u$ in $H^1(P)$, the matrices A_η also give the required lower bound for the volume energy. Indeed $A_\eta^{1/2}\nabla u_\eta \rightharpoonup \nabla u$ in $L^2(P)$: for each L^2 test vector z, dominated convergence gives $A_\eta^{1/2} z \to z$ strongly in $L^2(P)$. Weak lower semicontinuity of the squared norm yields

$$\int_P |\nabla u|^2 \leq \liminf_{\eta \downarrow 0} \int_P \nabla u_\eta \cdot A_\eta \nabla u_\eta.$$

Finally $J_\eta \to 1$ uniformly preserves the orthonormality of the limiting first j eigenfunctions. Applying the min–max principle to their j-dimensional span, exactly as in the thin-limit argument, gives the reverse eigenvalue inequality and proves convergence. □

Corollary 8.2 (Preservation of finitely many strict violations)**.** *Suppose a bounded open convex polytope P has strictly positive midpoint defects*

$$\lambda_j(P;\alpha_-) + \lambda_j(P;\alpha_+) - 2\lambda_j(P;(\alpha_- + \alpha_+)/2) > 0$$

for a finite collection of indices and parameter pairs with $0 < \alpha_- < \alpha_+$. Then one bounded C^∞ strictly convex domain has all the same strict midpoint violations.

Proof. Choose P_η as in the lemma. For every fixed index and parameter the eigenvalues converge, so every displayed defect converges to a strictly positive number. A single sufficiently small $\eta > 0$ works for the finite collection. □

Theorem 8.3 (Smooth strictly convex counterexamples in every dimension)**.** *For every $d \geq 2$ and $M \geq 2$, there are a bounded domain $D \subset \mathbb{R}^d$ with real-analytic boundary and strictly positive principal curvatures, and an $\alpha > 0$, such that*

$$\lambda_j(D;\alpha) + \lambda_j(D;3\alpha) > 2\lambda_j(D;2\alpha) \qquad (2 \leq j \leq M).$$

Proof. Choose a polytope from Corollary 7.1. There are only $3(M-1)$ eigenvalue evaluations to preserve. Apply Lemma 8.1 and choose a single sufficiently small smoothing parameter for all of them. The three Robin parameters remain fixed throughout this last approximation. □

9 Scaling and prescribed parameter values

Proposition 9.1 (Scaling preserves a strict defect)**.** *For every bounded Lipschitz domain D, $s > 0$, and $\alpha \geq 0$,*

$$\lambda_j(sD;\alpha) = s^{-2}\lambda_j(D;s\alpha). \tag{19}$$

Thus a strict concavity violation survives any homothety, with the three Robin parameters divided by s. In particular, the domains in this paper may separately be normalized to unit volume or unit diameter.

Proof. Under $x = sz$, the mass, gradient energy and boundary integral scale by s^d, s^{d-2}, s^{d-1}, respectively. Their Rayleigh quotient is s^{-2} times the quotient on D with parameter $s\alpha$. Apply min–max. Multiplication of the defect by $s^{-2} > 0$ preserves its sign. □

Corollary 9.2 (Any prescribed positive midpoint triple)**.** *Let $0 < a < b < c$ with $b = (a+c)/2$, and let $M \geq 2$. There is a bounded convex planar trapezoid D for which*

$$\lambda_j(D;a) + \lambda_j(D;c) > 2\lambda_j(D;b), \qquad 2 \leq j \leq M.$$

There is also such a domain with real-analytic strictly convex boundary.

Proof. Write $r = (c-a)/2$. The triangular-kernel identity on $[a,c]$ and Lemma 3.2 give

$$\delta^{-2}\big(\Lambda_{n+1}(\delta,a) + \Lambda_{n+1}(\delta,c) - 2\Lambda_{n+1}(\delta,b)\big) \longrightarrow r^2 q_n > 0$$

for $f = x$ and $n \geq 1$. The thin-domain argument therefore produces a fixed trapezoid $\Omega_{\varepsilon,\delta}$ with positive defects at $\varepsilon a, \varepsilon b, \varepsilon c$, simultaneously for $2 \leq j \leq M$. Set $D = \varepsilon\Omega_{\varepsilon,\delta}$ and use (19). Apply the smooth approximation lemma to the finite collection of strict defects for the final assertion. □

10 What is proved, and what the novelty audit establishes

10.1 The mathematical conclusion

Theorem 1.1 supplies a strict inequality in the reverse direction to the midpoint inequality for concavity. A single bounded convex domain and three strictly positive values of the same boundary parameter occur in that inequality. This is an unconditional disproof of the original Conjecture C, with no alteration of the geometric class or the quantity under discussion. The stronger conclusions concern any prescribed finite collection of excited eigenvalues, all ambient dimensions at least two, and real-analytic strictly convex boundaries.

The proof needs only standard variational spectral theory, Sobolev compactness and trace facts, and isolated-eigenvalue perturbation. The relevant special cases of domain convergence and boundary regularization have been proved explicitly here. General Robin form theory may be found in [5]; the perturbation framework is classical [4].

The thresholds δ_M and $\varepsilon_M(\delta)$ are existential thresholds whose existence is proved by the stated limits. We do not assign numerical values to them. This does not leave a conditional step in an existence theorem: the limiting coefficient is strictly positive and the defining convergence estimates hold for every sufficiently small parameter. Neither a finite computation nor a plot is used to replace such an estimate.

10.2 Related results and the limits of priority claims

The original conjecture and two later primary problem lists are cited in Section 1. We also checked the full text of Dai–Sun [6], which addresses Laugesen's Conjecture B concerning the ratio of two eigenvalues. Its thin-passage construction concerns a different geometric mechanism and a different target. The public AIM manuscript of Colbrook [7] proposes a counterexample to the diameter-normalized fundamental-gap conjecture using thin convex three-dimensional domains. It is relevant prior work for thin-domain spectral constructions, and does not claim the parameter-concavity result proved here.

The local predecessor concerning the first eigenvalue under domain dilation [9] also has a different target, parameter convention, and geometric construction. No theorem from that manuscript or from the VMRC corpus is required as an input to the present proof.

Searches through 31 October 2026 used the exact conjecture name and combinations of *Robin*, *second eigenvalue*, *concavity*, *positive parameter*, *convex*, *thin domain*, and *trapezoid*. The scope included the primary sources just listed and a duplicate search through the 95 TeX files of the specified corpus snapshot. No identical earlier resolution was located. This is a documented search result, not a proof of universal absence of prior literature.

The method combines established ingredients with the explicit affine-profile response. We do not claim that min–max, reduced-resolvent perturbation, thin-domain convergence, or smooth convex approximation is a historically new general principle. The asserted contribution is the exact positive coefficient, its simultaneous finite-mode consequence, and their application to the unchanged named conjecture.

Requested success condition	Evidence and status
Preexisting named problem	Laugesen's Conjecture C (2019), with independent 2020 and 2023 problem-list records.
Original problem preserved	The same ordinary Robin operator, bounded convex domain class, second eigenvalue, and range $\alpha > 0$.
Prior open status	Explicit in the cited records; no later identical resolution found in the stated search.
Unconditional complete argument	Theorem 1.1, the exact profile calculation, and the variational limit prove a strict reversed midpoint inequality.
More than a restatement of an old theorem	The displayed convex-domain violation is a new conclusion of the present argument; its component tools are classical.
No identical prior solution	None located. Historical firstness is not certified by the search, and is not assumed in the mathematical proof.

The paper does not assign itself an impact ranking relative to the corpus. Its significance can be evaluated from the exact preexisting statement answered and the scope of the proved extensions.

Acknowledgment and reproducibility

This research manuscript was prepared for the Keio research project with AI assistance. Separate mathematical derivations checked the signed coefficient, the boundary scaling, the min–max limits, and the quantifiers. These checks do not constitute independent human peer review or proof-assistant verification.

The source is a single self-contained `.tex` file: the diagram and bibliography are embedded. It compiles with standard pdfLaTeX packages, with no external figures, data, bibliography database, or shell execution. Two LaTeX passes resolve internal references. Appendix A records the elementary identities underlying the sign calculation and an optional independent numerical check; the proof itself does not depend on that check.

A Elementary checks of the signed coefficient

A.1 The correction equation

Put $k = n\pi$, $n \geq 1$, and write

$$A(x) = \frac{x^2 - x}{4k} - \frac{1}{4k^3}, \qquad B(x) = \frac{x - \frac{1}{2}}{4k^2}, \qquad y = \sqrt{2}(A\sin(kx) + B\cos(kx)).$$

The residual is

$$y'' + k^2 y = \sqrt{2}((A'' - 2kB')\sin(kx) + (2kA' + B'')\cos(kx)).$$

Here $A'' - 2kB' = 0$ and $2kA' + B'' = x - \frac{1}{2}$. Moreover

$$kA + B' = \frac{x^2 - x}{4},$$

which vanishes at both endpoints, while $\sin(kx)$ also vanishes there. This verifies the differential equation and both Neumann conditions in Lemma 4.1. Orthogonality follows from reflection about $x = 1/2$: $(x^2 - x)\sin(kx)\cos(kx)$ and $(x - \frac{1}{2})\cos^2(kx)$ both integrate to zero.

The three integration identities used in that lemma follow from $\sin(kx)\cos(kx) = \frac{1}{2}\sin(2kx)$ and $\cos^2(kx) = \frac{1}{2}(1+\cos(2kx))$. Integration by parts gives

$$\int_0^1 x\sin(2kx)\,dx = -\frac{1}{2k}, \qquad \int_0^1 (x^3-x^2)\sin(2kx)\,dx = \frac{3}{4k^3},$$

and

$$\int_0^1 x\cos^2(kx)\,dx = \frac{1}{4}, \qquad \int_0^1 x^2\cos^2(kx)\,dx = \frac{1}{6}+\frac{1}{4k^2}.$$

Substitution yields

$$\langle x\phi_n, y_n\rangle = \frac{3}{16k^4}+\frac{1}{8k^4}+\frac{1}{48k^2}+\frac{1}{8k^4} = \frac{1}{48k^2}+\frac{7}{16k^4}.$$

A.2 An independent positive lower bound for the second mode

The sign for $n=1$ can also be checked without summing the resolvent exactly. Let $a_m = \langle x\phi_1, \phi_m\rangle$. Elementary integration and Parseval give

$$a_0 = -\frac{2\sqrt{2}}{\pi^2}, \qquad a_1 = \frac{1}{2}, \qquad \sum_{m\neq 1} a_m^2 = \frac{1}{12}+\frac{1}{2\pi^2}.$$

Every denominator $\pi^2 - E_m$ for $m \geq 2$ is at most $-3\pi^2$. Hence

$$\begin{aligned}\frac{1}{2}\mathcal{Q}_1(x) &= \frac{a_0^2}{\pi^2} + \sum_{m\geq 2}\frac{a_m^2}{\pi^2 - E_m} \\ &\geq \frac{8}{\pi^6} - \frac{\frac{1}{12}+\frac{1}{2\pi^2}-\frac{8}{\pi^4}}{3\pi^2} = \frac{\frac{32}{\pi^4}-\frac{1}{12}-\frac{1}{2\pi^2}}{3\pi^2} > 0.\end{aligned}$$

For instance, the elementary bounds $9 < \pi^2 < 10$ make the last quantity greater than $163/27000$. This independently verifies the strict sign needed to refute Conjecture C.

A.3 Optional numerical replay

The following Python 3 program uses only the standard library. It compares the closed formula with a truncated cosine-series calculation. It is a diagnostic replay, not a premise of any theorem and not a numerical solution of the Robin eigenproblem. For $m \neq n \geq 1$,

$$\langle x\phi_n, \phi_m\rangle = I_{m-n} + I_{m+n}, \qquad I_r = \begin{cases}\frac{1}{2}, & r = 0,\\ ((-1)^r - 1)/(r\pi)^2, & r \neq 0.\end{cases}$$

The $m = 0$ coefficient is $\sqrt{2}I_n$.

```
from math import pi, sqrt

def I(r):
    if r == 0:
        return 0.5
    return (((-1) ** r) - 1) / ((r * pi) ** 2)

def response_series(n, cutoff=20000):
    E = (n * pi) ** 2
    total = 0.0
    for m in range(cutoff + 1):
        if m == n:
            continue
        if n == 0:
            coefficient = sqrt(2) * I(m)
```

```
        elif m == 0:
            coefficient = sqrt(2) * I(n)
        else:
            coefficient = I(m - n) + I(m + n)
        total += 2 * coefficient ** 2 / (E - (m * pi) ** 2)
    return total

def response_closed(n):
    if n == 0:
        return -1 / 60
    k = n * pi
    return (k * k + 21) / (24 * k ** 4)

for n in (0, 1, 2, 3, 5, 10, 30):
    exact = response_closed(n)
    replay = response_series(n)
    assert abs(exact - replay) < 1e-11
    assert (exact < 0) if n == 0 else (exact > 0)
    print(n, format(exact, ".12g"),
          format(abs(exact - replay), ".3g"))
```

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